

Package Insert

ANCARE SOLUTION FOR IV INFUSION 100MG/ML "

Ingredient:

Each mL contains active substance:

N-acetyl-L - cysteine.....100 mg

Acetylcysteine is the nonproprietary name for the N-acetyl derivative of the naturally occurring amino acid, L-cysteine (N-acetyl-L-cysteine, NAC).

Product Description:

Clear, colorless solution in colorless ampoule

R05CB01 acetylcysteine

Pharmacodynamics:

Paracetamol is metabolised in the liver, mainly by conjugation with glucuronide and sulphate. It is also metabolised by cytochrome P450 to form a reactive, potentially toxic metabolite. This metabolite is normally detoxified by conjugation with hepatic glutathione, to form non-toxic derivatives. In paracetamol overdose, the glucuronide and sulphate conjugation pathways are saturated, so that more of the toxic metabolite is formed. As hepatic glutathione stores are depleted, this toxic metabolite may bind to hepatocyte proteins, leading to liver cell damage and necrosis. Acetylcysteine is a sulphhydryl (SH) group donor, and may protect the liver from damage by restoring depleted hepatic-reduced glutathione levels, or by acting as an alternative substrate for conjugation with, and thus detoxification of, the toxic paracetamol metabolite.

Pharmacokinetics:

Acetylcysteine is the N-acetyl derivative of the naturally occurring amino acid, L-cysteine, and is deacetylated in the liver to cysteine, or oxidized to other metabolites such as N-acetylcystine or N,N-diacetylcystine. The parent compound and metabolites may be present in the plasma either free or protein bound. Renal clearance accounts for about 30% of total body clearance. Following intravenous administration, mean terminal half lives have been calculated to be 1.95 and 5.58 hours respectively for reduced and total acetylcysteine.

Drug-Drug Interactions:

No drug-drug interaction studies have been conducted.

Indication:

Acetylcysteine serves as an antidote of deliberate or accidental acetaminophen poisoning.

Dosage and Administration:

- Administration of this product should only be taken by physicians.
- Intravenous Injection: detoxification of acetaminophen poisoning

Diluted with dextrose 5% and infused IV. Initial dose: 150 mg/kg in 200 mL over 60 minutes. 2nd dose: 50 mg/kg in 500 mL over 4 hours. 3rd dose: 100 mg/kg in 1 L over 16 hours. Total dose: 300mg/kg in 20 hours.

Instructions for use:

1. Wash hands thoroughly with soap and water.
2. Check the dose of the medication.
3. Look to see if there is any liquid at the top of the ampoule. If there is, gently flick the ampoule with finger to get all the liquid into the bottom portion of the ampoule.
4. Hold the bottom of the ampoule in one hand. Clean the ampoule neck with an alcohol wipe using other hand. Then place the alcohol wipe around the neck of the ampoule and break it open by pressing the thumb against the neck of the ampoule.
5. To break, place thumb on the color line and snap back.
6. Tilt the ampoule down at a 45° angle. Insert the needle into the solution in the ampoule.
7. Draw up the medication by pulling back the plunger slowly and steadily until reach the dose.
8. Check the syringe for air bubbles. Hold it with the needle pointing upward. If there are air bubbles, tap finger against the barrel of the syringe to get the bubbles to the top. Slowly and carefully push the plunger up so that the bubbles are pushed out through the needle and see a drop of medication.
9. When there are no air bubbles, check the dose of the medication. If the dose is incorrect, repeat steps 5 through 8 until the right dose is drawn.

Warnings:

Hypersensitivity Reactions Serious acute hypersensitivity reactions during acetylcysteine administration including rash, hypotension, wheezing, and/or shortness of breath, have been observed in patients receiving intravenous acetylcysteine for paracetamol overdose and occurred soon after initiation of the infusion (see Adverse

Effects/Undesirable Effects). If a severe hypersensitivity reaction occurs, immediately stop the infusion of acetylcysteine and initiate appropriate treatment.

Acute flushing and erythema of the skin may occur in patients receiving acetylcysteine intravenously. These reactions usually occur 15 to 60 minutes after initiating the infusion and often resolve spontaneously despite continued infusion of acetylcysteine. If a reaction to acetylcysteine involves more than simply flushing and erythema of the skin, it should be treated as a hypersensitivity reaction.

Management of less severe hypersensitivity reactions should be based upon the severity of the reaction and include temporary interruption of the infusion and/or administration of antihistaminic drugs. The acetylcysteine infusion may be carefully restarted after treatment of the hypersensitivity symptoms has been initiated; however, if the hypersensitivity reaction returns upon re-initiation of treatment or increases in severity, acetylcysteine should be discontinued and alternative patient management should be considered.

Precautions:

Ancare should be used with caution in patients with asthma, or where there is a history of bronchospasm. The total volume administered should be adjusted for patients less than 40 kg and for those requiring fluid restriction. To avoid fluid overload, the volume of 5% dextrose should be reduced as needed. If volume is not adjusted fluid overload can occur, potentially resulting in hyponatremia, seizure, and death.

Interaction with Other Medicines:

Drug stability and safety of acetylcysteine when mixed with other drugs have not been established.

Side Effects:

Immune System Disorders: Anaphylactic/anaphylactoid reaction

Skin and Subcutaneous Tissue Disorders: Severe cutaneous adverse reactions (SCAR) e.g. erythema multiforme, Stevens-Johnson syndrome (SJS) and toxic epidermal necrolysis (TEN). In most of these cases reported at least one other drug was administered at the same time, which may have possibly enhanced the described mucocutaneous effects.

Effects on Ability to Drive and Use Machine:

No information on this is known.

Usage during Pregnancy and Lactation:**Pregnancy:**

Category B2 There was no evidence of teratogenicity in limited studies in rats and rabbits following administration of Acetylcysteine during the period of gestation at doses up to 1.2 times the maximum clinical dose, on a body surface area basis. There are no well-controlled studies in pregnant women but experience does not include any positive evidence of adverse effects to the foetus.

Lactation:

There was no evidence of adverse effects in a limited study in rats following administration of acetylcysteine during late gestation and lactation at 60% of the maximum clinical dose, on a body surface area basis. It is not known whether acetylcysteine and/or its metabolites are excreted in milk. There are no data on the use of acetylcysteine in lactating women and therefore breastfeeding is not recommended during treatment.

Symptoms and Treatment of Overdose:

Single intravenous doses of acetylcysteine at 1000 mg/kg in mice, 2445 mg/kg in rats, 1500 mg/kg in guinea pigs, 1200 mg/kg in rabbits and 500 mg/kg in dogs were lethal. Symptoms of acute toxicity were ataxia, hypoactivity, labored respiration, cyanosis, loss of righting reflex and convulsions. There is no specific treatment. General supportive measures should be carried out. It has been suggested that generalised reactions to acetylcysteine can be treated with intravenous injection of an antihistamine, and infusion of acetylcysteine should be temporarily stopped but can be restarted at a slower rate without further reaction.

Contraindication:

It is contraindicated in patients with hypersensitivity or previous anaphylactoid reactions to acetylcysteine or any components in the preparation.

Storage & shelf life for reconstitution solution:

The injection is stable during the test period in the following temperatures and conditions.

- Finished-product : 5% Dextrose = 1 : 1
at 25°C ; test period : 16hr

- Finished-product 150mg/kg in 200 mL 5% Dextrose at 25°C ; test period : 1hr
- Finished-product 50mg/kg in 500 mL 5% Dextrose at 25°C ; test period : 16hr

Storage:

Store in a cool place below 30°C .

Packaging:

The product is available in 3mL, 5mL, and 10 mL glass ampoule. 50 ampoules are packed in one box. Not all pack sizes will be marketed.

Manufacturer:

NANG KUANG PHARMACEUTICAL CO., LTD.

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Date of Revision:

February 2025